THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS TWO-CHAPTER TEST SET D

Chapters 3 & 4: **The World of Numbers, Exploring Algebraic Identities** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The value of $(-12) \times 5$ is: 1

(a) -58 **(b)** 59 **(c)** -60 **(d)** 60

2. The set of all rational numbers is denoted by: 1

(a) $\mathbb{N}$ **(b)** $\mathbb{R}$ **(c)** $\mathbb{Q}$ **(d)** $\mathbb{Z}$

3. 0.999… ($0.\overline{9}$) is equal to: 1

(a) 1 **(b)** 0.9

(c) 0.99 **(d)** slightly less than 1

4. The distance between -4 and 3 on the number line is: 1

(a) 12 **(b)** 1 **(c)** 7 **(d)** −7

5. The block 142857 repeats in the decimal of: 1

(a) $\frac{1}{3}$ **(b)** $\frac{1}{7}$ **(c)** $\frac{1}{9}$ **(d)** $\frac{1}{11}$

6. Factorise $x^2 - 5x - 36$: 1

(a) $(x-4)(x+9)$ **(b)** $(x-9)(x+4)$ **(c)** $(x+4)(x+9)$ **(d)** $(x-9)(x-4)$

7. $(2n+7)(2n-7)$ equals: 1

(a) $4n^2 - 49$ **(b)** $4n^2 - 45$ **(c)** $4n^2 + 28n + 49$ **(d)** $4n^2 - 28n + 49$

8. The coefficient of ab in $(a+b+c)^2$ is: 1

(a) 1 **(b)** 3 **(c)** 4 **(d)** 2

9. If $a = 10$ and $b = 2$, then $(a+b)^2 - (a^2 + b^2)$ equals: 1

(a) 0 **(b)** 40 **(c)** 144 **(d)** 20

10. $(x+y+z)^2$ contains how many terms when fully expanded (no like terms)? 1

(a) 4 **(b)** 9 **(c)** 3 **(d)** 6

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the distance between $-\frac{1}{2}$ and $\frac{5}{2}$ on the number line. 2

12. A rational number in lowest form has denominator $2^3 \times 5$. How many decimal places does its expansion have? Why? 2

13. Find 55^2 and 65^2 using $a^2 = (a+b)(a-b) + b^2$ with $b = 5$. 2

14. Show that the identity $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$ holds for $a = 2,\ b = 1$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the lengths of the first three hypotenuses in the square-root spiral starting with a right triangle of legs 1, 1. 3

16. Factorise $6x^2 + 7x + 2$ by splitting the middle term. 3

17. Find the decimal expansion of $\frac{1}{7}$ and identify its repeating block. What is special about it? 3

OR

Show that the average $\frac{a+b}{2}$ of two rational numbers $a < b$ lies between them. 3

18. Use $(x-y)(x^2+xy+y^2) = x^3 - y^3$ to factorise $27a^3 - 8b^3$. 3

OR

If $x + \frac{1}{x} = 3$, find $x^2 + \frac{1}{x^2}$ and $\left(x - \frac{1}{x}\right)^2$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Show that $0.\overline{142857} = \frac{1}{7}$ and verify the cyclic property by computing 2×142857 and 3×142857. 5

OR

(i) Find the distance between -4 and 3, and between $-\frac{5}{2}$ and $\frac{7}{2}$, on the number line. 5
(ii) Write $\left|-\frac{7}{9}\right|$, $|0|$ and $\left|\frac{5}{3}\right|$.
(iii) Between which two consecutive integers does $\frac{13}{4}$ lie?

20. Case study: Explore $(x-y)(x^2+xy+y^2)$ and $(x+y)(x^2-xy+y^2)$. 5
(i) Multiply and state the identities.
(ii) Factorise $27p^3 + 8q^3$ and $64 - a^3$.

OR

If $x + \frac{1}{x} = 5$, find $x^2 + \frac{1}{x^2}$ and $x^3 + \frac{1}{x^3}$. 5

END OF QUESTION PAPER — Total: 40 Marks

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